

Shadow Product Proposal v1.0

Local-only AI Mentor and Compliance Council for Regulated Knowledge Work

Alex Xiaoyu Ji

2026-06-17 (v1.0 — canonical product proposal)

Shadow

One Engine, Three Smart-Glass Clients, Five Persona Packs

Canonical product proposal v1.0 — supersedes all prior framing of council-diff, ECC banking-quest, ECC trading-quest as separate products. Going forward, *all four projects share one product narrative and one architectural diagram.*

1. Executive Summary

Shadow is the **on-device AI council** that runs locally on the user's laptop and three classes of smart glasses. The engine is the same for every user. The smart-glass client adapts to *what the user is doing right now*, and the persona pack adapts to *who the user is*.

The result is a **regulator-grade, local-only, hardware-flexible** AI mentor and compliance second-opinion layer that no Cloud-only competitor (Glean, Microsoft Copilot, Goldman GS-AI, JPM LLM Suite) can match, because the moat is the *on-device + smart-glass + persona-aware* composition, not any single one of these.

Same engine. Three glass clients. Five persona packs. Fifteen recognized scenarios. One regulated-grade contract per bank.

2. The Problem — Regulated Knowledge Work Has Five Stuck Personas

Across mid-tier US banks (\$5B–\$50B AUM), five distinct roles each face a stuck workflow:

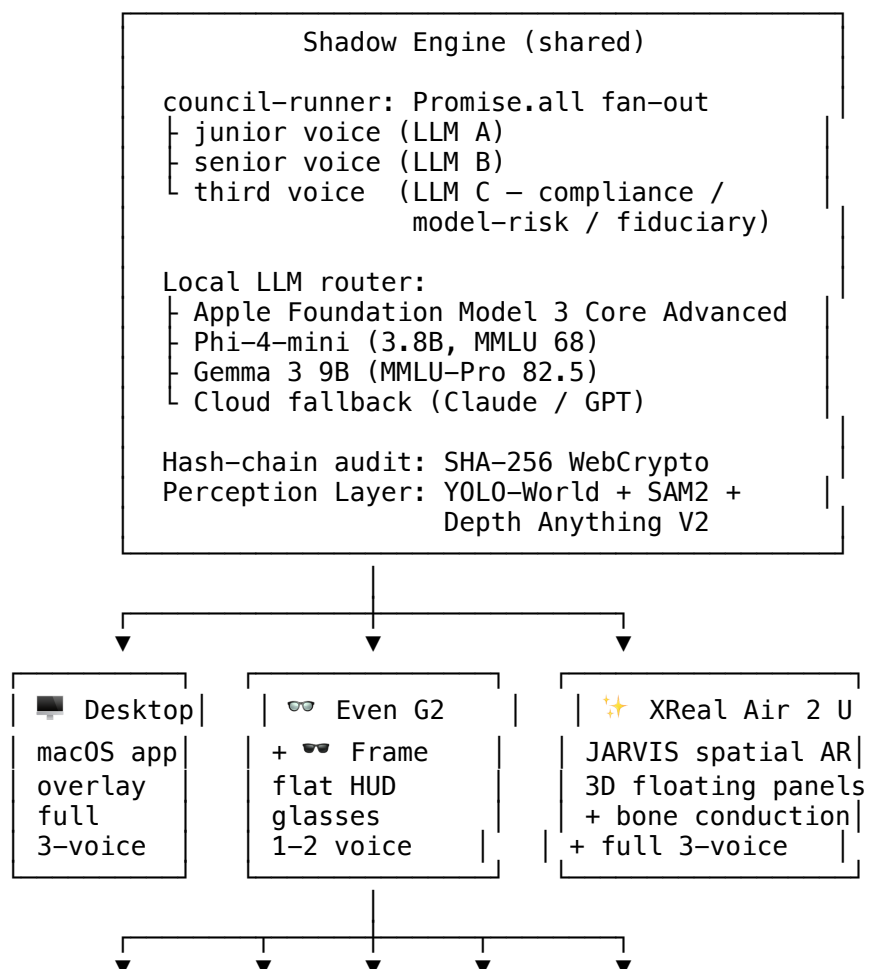
- **Compliance officers** rubber-stamp by 3pm because every loan file feels template-identical, and OCC examiners demand contemporaneous evidence of contrarian challenge that 2D Word memos cannot produce.

- **Quantitative analysts and data scientists** know their model PSI tripped on a feature drift, but the next question — *which feature, on which sub-population, contributed how much* — requires writing a SHAP-attribution notebook from scratch, every time.
- **Software engineers** integrate ML services into the credit-decisioning critical path without spatial awareness of where Fair Lending data needs to be redacted, where service boundaries should be async, or where covered attributes need explicit Reg B logging.
- **Traders and portfolio managers** watch a 2D correlation heatmap that hides the time-varying regime shift behavior — the very thing that wipes out portfolios in stress.
- **Wealth advisors** sit across a table from HNW clients while pulling out a laptop or a tablet to look up Reg BI suitability data, breaking eye contact and customer trust precisely when the trust matters most.

A single 2D dashboard cannot serve all five. Neither can a single XR form factor. Neither can a generic LLM chat box behind a compliance gateway.

What can serve all five is **one engine** that emits the same JSON verdict, **three glass clients** that render the verdict at the right cognitive density for the moment, and **five persona packs** that load role-specific voices, jargon, regulations, and historical Brier audits.

3. The Architecture — One Engine, Three Clients, Five Packs



 Compl.	 Quant	 Eng	 Trader	 Advisor
pack	pack	pack	pack	pack
\$1,800/y	\$2,400	\$1,500	\$2,400	\$2,000
per seat	/seat	/seat	/seat	/seat

Same JSON verdict. The engine emits the same data: 3 voice verdicts with rationale, a follow-up question, a recognized-context tag, and a hash-chain entry. What changes is *which client renders it* (desktop vs glasses vs XReal AR) and *which persona pack supplies the voices and the domain knowledge*.

4. The Persona × Device Matrix — What Each Combination Produces

This is the heart of the product. Each cell describes the **concrete output the user sees** in that combination.

Compliance Officer

	 Desktop	 Even G2 (no camera)	 Brilliant Frame (camera, local)	 XReal Air 2 Ultra (spatial AR, camera, local)
Best fit?	✓ Daily review of 30–50 loans	✓ Primary during customer meetings	△ Camera = customer trust issue	△ Overkill for paperwork
What user sees	Right-side panel: 3-voice rationale, follow-up question, fair-lending flag table	Single contrarian whisper: “DTI 47% > Policy 4.3 ceiling. Why approve?” in green monocular HUD	Color HUD with 2 voices + flag icons on borrower attributes	Not the primary device; reserved for quarterly exam-prep deep dives where compliance officer walks past 30 past denials in a calibration corridor
Audit output	Full hash-chained transcript: text response logged + timestamp	Bone-conduction whisper response transcribed to chain	Same as G2 but with screen-region tag	Gaze-dwell timeline added to chain (eye tracking proves the officer actually viewed the disparate-impact flag)

Quant / Data Scientist

	 Desktop	 Even G2	 Brilliant Frame	 ✨ XReal Air 2 Ultra
Best fit?	✅ IDE work, notebook attribution	⚠️ Too low-bandwidth for SHAP / drift	✅ Reviewing internal model docs	✅ Primary for AML graph + fraud manifold
What user sees	Drift attribution panel: feature × sub-population × contribution table next to the Jupyter notebook	Single whisper if PSI tripped: “3 <i>features drifted</i> . <i>Sub-population: consumer-discretionary.</i> ”	Color HUD: 2 voices on top of a model-card PDF + “run SHAP” suggestion	JARVIS mode: 3 floating panels anchored to physical desk. (1) 3D fraud feature-space UMAP — walk around to find misclassified cluster. (2) Counterparty network 5K-node walk-through. (3) Drift attribution timeline. Bone-conduction whispers model-risk reviewer’s challenge.
Audit output	Notebook cell + SHAP output hash-chained	Voice transcript	Same	Gaze + walking trajectory + which floating panel was inspected — all hash-chained for SR 11-7 effective challenge evidence

Software Engineer

	 Desktop	 Even G2	 Brilliant Frame	 ✨ XReal Air 2 Ultra
Best fit?	✅ Primary in IDE	⚠️ Too low-bandwidth	✅ Internal doc / runbook reading	⚠️ Overkill
What user sees	Right-side panel + inline code	Single whisper for emergency Slack	Color HUD on top of a Confluence	Not primary; XReal is for visual data

	 Desktop	 Even G2	 Brilliant Frame	 XReal Air 2 Ultra
	annotations: “this endpoint takes regulated PII, log to Reg B store before returning”	alerts: “ <i>deploy on hold — security scan flagged your PR</i> ”	runbook, highlights the section the engineer is reading + suggests adjacent runbooks	work, not IDE work
Audit output	Code-change-to-decision link in hash chain	n/a	Doc-section-viewed timeline	n/a

Trader / Portfolio Manager

	 Desktop	 Even G2	 Brilliant Frame	 XReal Air 2 Ultra
Best fit?	✓ Daily Bloomberg work	✓ During client / desk-team meetings	⚠ Camera in IB = MNPI risk	✓ Primary at own desk for risk-surface + tail-risk
What user sees	Right-side panel: 3 trader voices (Buffett value / Druckenmiller macro / Taleb tail) + bias constellation snapshot	Single whisper during desk meetings: “ <i>counterparty 17 is on watchlist</i> ”	Not the primary use case	JARVIS mode: 3 floating panels — (1) 3D risk surface (VaR + ES + concentration on 3 axes) — walk around to find the cliff. (2) Bias constellation: 30 past trades clustered, walk to inspect biases. (3) Correlation surface across regimes — drag time slider, watch 2007→2009 deformation. Bone-conduction council whispers.
Audit output	Brier-audited trade thesis log	Voice transcript	n/a	Gaze + walking + decision-time log

🖥️ Desktop

🕶️ Even G2

🕶️ Brilliant Frame

✨ XReal Air 2
Ultra

— full forensic
replay for post-
mortem

👛 Wealth Advisor

🖥️ Desktop

🕶️ Even G2

🕶️ Brilliant Frame

✨ XReal Air 2
Ultra

Best fit?	✅ Portfolio review at desk	✅ Primary in client meetings Single whisper: <i>“client mentioned 401k rollover — Reg BI suitability checklist surfacing”</i> + adverse-action-safe phrasing for the IPS conflict	⚠️ Camera kills client trust	⚠️ Overkill
What user sees	Right-side panel: 3 advisor voices + Reg BI suitability flag + IPS gap analysis	n/a		Not primary; could be used for quarterly book review
Audit output	IPS-aligned recommendation log	Conversation transcript with timestamped Reg BI checks	n/a	n/a

Summary — what each cell *produces*

Desktop = full bandwidth, every voice, every detail. The universal fallback. Where heavy review work actually gets done.

Even G2 = single contrarian whisper + transcription. The customer-facing primary. Where adverse-action evidence is captured during conversation without breaking trust.

Brilliant Frame = color HUD + 2 voices + camera context. The internal-doc reading and runbook-walk-through device. Where junior staff get explainer-style HUD prompts on real-world documents.

XReal Air 2 Ultra = JARVIS spatial AR + 3 floating panels + bone conduction. The power-user device. Where 3D data work (AML graph, fraud manifold, risk surface, bias constellation, correlation regime) becomes physically inspectable.

5. Local-only Mode — The Moat

Every glass client + the desktop overlay has two modes:

- **Cloud mode:** screen capture → Anthropic / OpenAI / Gemini → response. For non-confidential training data, public market signals, educational content. \$50/seat/month consumer SKU.
- **Local mode** (the moat): screen capture → on-device Apple Foundation Model 3 Core Advanced, Phi-4-mini, Gemma 3 9B, or Mistral Small 3 → response. **No data leaves the device.** Frame discarded in 200ms. Voice transcripts redacted of PII before hash-chain commit. Hash chain entry only includes the *response*, never the source frame.

What local mode solves

-  Data residency (GLBA / HIPAA / GDPR Article 25 data minimization)
-  Data upload (no packet inspection, no third-party DPA chain)
-  Vendor concentration risk (no Anthropic / OpenAI dependency)

What local mode does NOT solve

-  “Camera exists” liability under 2-party consent state law (California, Illinois, Massachusetts, Pennsylvania) — solved by selecting **Even G2 (no camera)** for the customer-facing cells of the matrix
-  Customer trust during meetings — solved by Even G2 device choice
-  E&O insurance and HR policy — solved by Even G2 + Desktop, and by limiting camera-equipped glasses to no-customer-present scenarios

The cross-tier audit chain works in local mode

Every persona × device combination contributes to the same SHA-256 hash chain. The chain stores:

- Recognized context (screen / scene tag)
- Question surfaced by the council
- Voice transcript (challenge text + user verbal or gestural response)
- Gaze-dwell timeline (Vision Pro / XReal eye tracking)
- Hand-gesture confirmation (approve / block / escalate)

Exportable to OCC, CFPB, FINRA, SEC, EU AI Act Article 14 audit submission formats. All generated on-device. Zero cloud dependency.

6. Pricing Model

Shadow Banking · Enterprise License

Base engine	\$120,000/year (no seat cap)	
— Persona Packs (per seat per year) —		
 Compliance	\$1,800/seat	
 Quant / Data Scientist	\$2,400/seat	
 Engineering	\$1,500/seat	
 Trader	\$2,400/seat	
 Wealth Advisor	\$2,000/seat	
— Hardware (sold separately) —		
 Even G2	\$599	one-time per user
 Brilliant Frame	\$349	one-time per user
 XReal Air 2 Ultra	\$699	one-time per user

Sample mid-tier bank deployment (1,000 employees):

Persona	Seats	Pack \$/yr	Subtotal
Compliance	200	\$1,800	\$360K
Quant + Data Scientist	50	\$2,400	\$120K
Software Engineering	100	\$1,500	\$150K
Trader	30	\$2,400	\$72K
Wealth Advisor	80	\$2,000	\$160K
Base engine	—	—	\$120K
Total annual contract value 460 seats —			\$982,000

3-year SOM target: 5 mid-tier banks × \$982K/yr = \$4.9M ARR. Series A scale.

7. Hardware Procurement Tiers (employer-paid)

- **Tier 0 — no glasses:** Desktop only, every employee. \$0 hardware cost.
- **Tier 1 — G2 for customer-facing roles:** 280 G2 units × \$599 = **\$168K one-time**. Covers all Compliance + Wealth Advisor.
- **Tier 2 — XReal Air 2 Ultra for power users:** 80 XReal units × \$699 = **\$56K one-time**. Covers all Quant + Trader.
- **Tier 3 — Brilliant Frame for engineers:** 100 Frame units × \$349 = **\$35K one-time**. Optional, only if engineering team requests.

Total Tier 1+2+3 hardware: ~\$260K one-time. Inside the typical mid-tier bank discretionary tech budget.

8. Roadmap

Quarter	Milestone
2026 Q3	macOS Desktop app (native ScreenCaptureKit + local Phi-4-mini, Gemma 3 9B). First 3 persona packs (Compliance / Quant / Trader). Hash-chain audit export. Browser demo (live now).
2026 Q4	Even G2 SDK integration (Conversate audio + bone conduction). 5 persona packs complete. SOC 2 Type 1 in progress. 5 mid-tier wealth-mgmt firm cold-email round.
2027 Q1	First paid pilot — 1 mid-tier bank, Compliance pack, Desktop + Even G2. SOC 2 Type 1 issued.
2027 Q2	Brilliant Frame integration (color HUD). EU AI Act high-risk obligations live (2026-08-01 entry-into-force) — Shadow's local-only audit positioning aligns with the August deadline.
2027 Q3	XReal Air 2 Ultra NRSDK integration (JARVIS spatial AR mode). First Quant + Trader pilots. ICAIF 2027 paper submission.
2027 Q4	Series A fundraising. 5 paying mid-tier bank contracts. \$4-5M ARR.
2028	Apple Vision Pro 2 (rumored binocular + cheaper) — add as Tier 4 device. Mainstream consumer-grade glasses. Expand from mid-tier to top-tier banks.

9. Defensibility

Against Glean / Microsoft Copilot / Notion AI: they are cloud + corpus-search; we are local + screen-aware + persona-aware.

Against Goldman GS-AI / JPM LLM Suite / Citi Stylus / Morgan Stanley AI@Morgan: they are top-10 internal builds; we sell to mid-tier banks that lack the engineering bench.

Against Cluely (\$5.3M raised, “cheat on everything” consumer overlay): closest UX competitor but consumer-positioned, brand cannot enter compliance-sensitive enterprises, ARR claims inflated and disputed March 2026.

Against future Meta / Apple glasses: hardware-neutral architecture means we add new device clients as they ship. Hardware refresh ≠ product reset.

Against academic-only competitors (ECC research repo, university spin-offs): founder has shipped 41 tests green across the council-runner stack + 18 tests on retry primitives +

10. Founder Credibility Wedge

- **Embodied Compliance Council** research with Dr. Henry Ngo (Yeshiva, banking compliance vertical) — EU AI Act + ECOA / Reg B framing locked through accepted academic deliverable
 - **Phase 4 EU AI Act + ECOA evaluation design** doc (docs/phase-4-eu-ai-act-evaluation-design.md)
 - **council-runner v0.1.0** shipped 2026-06-17, 41 tests green, Vercel-deployable, Apache 2.0
 - **Bilingual EN/中文** ship discipline opens secondary China-side market (CITIC, Haitong, HuaTai analyst classes) post-2027
 - **Solo Founder OS** 11-agent open-source stack — public proof of operating discipline
-

11. The One-Sentence Pitch

Shadow is the on-device AI council and audit chain that follows the user across desktop, smart glasses, and spatial AR, deploying role-specific voices for compliance officers, quants, engineers, traders, and wealth advisors at mid-tier US banks — never uploading data, never breaking customer trust, and producing the contemporaneous evidence that regulators are about to demand.

Appendix — Tech Stack Snapshot (2026-06-17)

Layer	Stack	Status
Council orchestration	council-runner (TypeScript, Promise.all, weighted aggregation)	Live, 23 tests green
Voice prompts	council-voices (10 banking + trading personas, YAML frontmatter, strict-JSON output)	Live
Risk math	orallexa-risk (VaR, ES, factor exposures, beta, concentration, sector, correlation)	Live, 11 tests green
Perception (screen-region)	embodied-perception (YOLO-World, SAM2, Depth Anything V2 protocols + Mocks + SceneGrounder)	Live, 7 tests green

Layer	Stack	Status
Audit	WebCrypto SHA-256 hash chain	Live, 4 tests green
Local LLM	Phi-4-mini, Gemma 3 9B, Apple Foundation Model 3 Core All available 2026 Advanced, Mistral Small 3	
Cloud LLM	Anthropic Claude Opus 4.7 (1M context), GPT-5, Gemini 3 Pro	All available
Desktop client	Native macOS ScreenCaptureKit + AppKit overlay	In design, Q3 2026 ship
Glasses client (G2)	Even Realities Companion SDK + Conversate	Q4 2026
Glasses client (Frame)	Brilliant Labs Open SDK	Q2 2027
Glasses client (XReal)	NRSDK + WebXR + Three.js	Q3 2027
Bone-conduction audio	Built-in to G2 + AirPods Pro 2 fallback for Frame and XReal	n/a

Canonical product proposal v1.0 — 2026-06-17. All prior project framing (council-diff as standalone OSS, ECC banking-quest as separate WebXR app, ECC trading-quest as separate app, Shadow as separate Phase 3) is hereby reconciled under this single product narrative. Going forward, every external communication uses this matrix.